
YohPal Live Video Upload & Editing Suite — Current Codebase Assessment

Daniel Macharia <danielmacharia43@gmail.com>
To: <daniel@ics.ac.ke>

Tue, 14 Jul at 11:23

YohPal Live Video Upload & Editing Suite — Current Codebase Assessment

I reviewed the available YohPal Live codebase diagnostic reports. The reports represent different repository snapshots between June and July 2026, so the findings below distinguish between **confirmed implementation**, **partially implemented capability**, and **features that still require runtime proof against the latest immutable commit**.

Executive assessment

YohPal currently has a **working upload pipeline and a meaningful AI publishing assistant**, but it does **not yet have a competitive native video-editing studio** comparable to what users expect from TikTok, Instagram Reels, CapCut or YouTube Shorts.

The strongest existing capability is:

Upload a gallery video → send it to Firebase Storage → create the video record → request AI-generated publishing intelligence → apply selected AI recommendations → publish into the feed.

The weakest area is:

Actual manipulation of the media itself.

Most current AI features generate text, scores or recommendations. They do not yet physically edit and render the video.

1. Functionalities currently available

A. Creator entry point

The Create hub currently provides access to:

- Video upload
- Go Live
- AI Editor
- YCIOS creator workspace
- Other creation-related cards

The July readiness review confirms that the Create hub's upload route resolves to a real screen. It also notes that some other Create hub cards were mislabeled duplicates, but the upload entry itself is functional.

YohPal Live — Production Readiness Review.pdf

Current status

Available

B. Gallery video selection

UploadScreen opens the device gallery through `image_picker`.

The creator can select a pre-recorded video stored on the phone.

YohPal Live — Production Readiness Review.pdf

Current status

Available

Current limitation

There is no confirmed option to:

- Record a new short video from the YohPal camera
- Switch front/rear camera
- Pause and resume recording
- Record multiple clips
- Set a recording countdown
- Select recording speed
- Use recording filters
- Use hands-free recording

The UI reportedly promises "Record or upload," but only gallery selection was found in the actual upload screen.

YohPal Live — Production Readiness Review.pdf

C. File validation

The current upload controller validates:

- File extension
- Maximum file size of approximately 250 MB

Current status

Available

Missing validation

The audit did not find proper validation for:

- Video duration
- Aspect ratio
- Resolution
- Codec
- Frame rate
- Corrupt media
- Audio track availability
- Orientation
- Minimum quality
- Copyright risk
- Duplicate upload detection

YohPal Live — Production Readiness Review.pdf

D. Firebase Storage upload

The current flow uploads selected video files to Firebase Storage.

Upload progress is connected to Firebase Storage `UploadTask.snapshotEvents`, meaning progress is real rather than simulated.

YohPal Live — Production Readiness Review.pdf

Current status

Available

Current UI capability

- Upload progress can be displayed.
- The uploaded asset is stored remotely.
- A video document is created after or during the upload workflow.

Missing reliability capabilities

- No connectivity pre-check before beginning a large upload
- No user-facing upload cancellation
- No retry button after failure
- No resumable upload recovery after application restart
- No queue for offline publishing
- No confirmed background-upload survival
- No upload speed or remaining-time estimate

The readiness review specifically found that `connectivity_plus` was present but unused in the upload journey, while the underlying Firebase task supported cancellation that had not been exposed to the user.

E. Firestore video-document creation

The upload system creates a Firestore video document.

Later pipeline fixes added or corrected:

- `hlsUrl`
- `videoUrl`
- `ownerId`
- `userId`
- Server-side `createdAt`
- `status`
- `visibility`
- Views
- Likes
- Engagement score
- HLS manifest fields

Earlier defects caused uploaded videos to remain invisible because the Cloud Function and Flutter model used different field names, `ownerId` was missing, dates were stored as strings, and the playable check incorrectly required an `.m3u8` extension. Those defects were reported as corrected.

YohPal Mobile Flutter — Gap Report v10 — Video Pipeline Fixed + Chat Bugs Found.pdf

Current status

Implemented, but current immutable-commit proof is required

F. Public-feed publication

The upload flow currently creates videos with states such as:

```
status: upload_pending
visibility: public
```

This means the upload can enter the public processing/publication path.

Important weakness

The AI workflow does not currently enforce publication readiness. The July audit found that the video could become publicly visible regardless of whether the creator completed the AI checklist.

YohPal Live — Production Readiness Review.pdf

Current status

Available, but governance is weak

2. Existing AI Creator Studio functionality

The AI Creator Studio is the most advanced part of the editing suite.

A. AI job infrastructure

The codebase contains:

- `AIVideoJob` model
- `AIVideoRepository`
- Job creation
- Active-job watching
- User-job history
- Cloud Function processing
- Job state lifecycle
- Audit events
- Per-user rate limiting

The reports describe a lifecycle similar to:

```
queued
→ processing
```

→ completed
or failed

The backend also records audit events such as:

- AI job started
- AI job completed
- AI job failed

The reported daily rate limit is 50 AI jobs per user.

Current status

Implemented

B. AI caption generation

The system can generate suggested captions for an uploaded video.

The creator can view and apply the generated caption to the video document.

Current status

Implemented in the AI publishing workflow

Important distinction

This appears to mean **post caption/copy generation**, not necessarily burned-in subtitle generation synchronized to spoken words.

C. AI hook generation

The AI system can suggest opening hooks intended to improve initial viewer retention.

Examples of possible output:

- Opening sentence
- First-three-second idea
- Curiosity hook
- Question hook
- Emotional hook

Current status

Implemented as generated recommendations

Limitation

There is no confirmed automatic replacement or editing of the actual opening video frames.

D. AI hashtag generation

The Creator Studio supports hashtag-generation jobs and can apply selected hashtags to the video record.

YohPal Mobile Flutter v3 — Gap Report v3.pdf

Current status

Implemented

E. AI viral score

The AI processor can return:

- Viral score
- Explanation
- Recommended improvements

The publishing repository can apply the viral score to the video document.

Current status

Implemented as an advisory score

Limitation

There is no evidence that the viral score is trained or calibrated using YohPal's real performance data.

At present it should be marketed as:

"AI content-readiness guidance"

rather than:

"Guaranteed viral prediction."

F. AI thumbnail ideas

The Creator Studio includes:

- Thumbnail-idea generation
- Thumbnail idea model
- Tap-to-select interface
- Application of the selected idea to the publishing record

YohPal Mobile Flutter v3 — Gap Report v3.pdf

Current status

Implemented as an idea selector

Limitation

There is no strong evidence that it:

- Extracts frames from the video
- Generates an actual image
- Adds text overlays to the image
- Renders thumbnail variants
- Performs face or expression selection
- Tests thumbnails against predicted click-through rate

It currently appears closer to a **thumbnail concept generator** than a complete thumbnail-production tool.

G. AI translation

An AI translation service exists and was reported as part of the six supported AI job types.

Current status

Backend/service implemented

Integration warning

The later production-readiness review described `ai_translation_service.dart` as existing but not wired into the live creator UI/controller.

YohPal Live — Production Readiness Review.pdf

Therefore:

Service exists; user-facing availability requires verification.

H. AI trim suggestions

`AiTrimService` and a `trim_suggestions` AI job were reported as implemented.

YohPal Mobile Flutter v3 — Gap Report v3.pdf

The AI may recommend:

- Sections to remove
- Suggested start time
- Suggested end time
- Slow or redundant moments

- Stronger opening segment

Current status

Suggestion generation exists

Critical limitation

There is no evidence that YohPal currently:

- Moves timeline handles
- Executes the trim locally
- Sends the trim to FFmpeg
- Produces a newly rendered video
- Allows the user to preview before/after
- Supports non-destructive edit history

This is **AI trim advice**, not yet a full auto-trim editor.

I. AI publication checklist

The current system reportedly has a five-item checklist and a “Mark Ready for Publishing” action.

The publishing repository supports actions such as:

- Apply caption
- Apply hashtags
- Apply viral score
- Apply thumbnail idea
- Mark AI-ready for publishing

YohPal Mobile Flutter v3 — Gap Report v3.pdf

Current status

Implemented in UI and Firestore state

Critical problem

The July audit found that the checklist does not actually block publication. The `aiPublishStatus` value is written, but the public feed and upload pipeline do not reliably enforce it.

YohPal Live — Production Readiness Review.pdf

This makes the checklist largely advisory.

3. Current upload and editing capability matrix

Functionality	Current position
Gallery selection	Available
In-app camera recording	Missing
File-extension validation	Available
File-size validation	Available
Duration validation	Missing
Firebase Storage upload	Available

Real upload-progress updates	Available
Cancel upload	Missing
Retry failed upload	Missing
Background/resumable upload	Missing
Connectivity pre-check	Missing
Firestore video record	Available
Processing status	Available
Public visibility	Available
HLS/ABR production transcoding	Evidence conflict; runtime proof required
AI captions	Available
AI hooks	Available
AI hashtags	Available
AI viral score	Available
AI thumbnail ideas	Available
AI translation	Service present; UI wiring uncertain
AI trim suggestions	Available as recommendations
Actual timeline trimming	Missing
Split/merge clips	Missing
Crop/rotate	Missing
Speed control	Missing

Filters/LUTs	Missing
Beauty enhancement	Missing
Background removal	Missing
Music library	Missing
Beat synchronization	Missing
Voice-over recording	Missing
Sound effects	Missing
Text overlays	Missing
Stickers/GIFs	Missing
Auto subtitles on video	Not proven
Transitions	Missing
Duet/stitch/remix	Missing
Green screen	Missing
Templates	Missing
Draft persistence	Not sufficiently proven
AI publish checklist	Present but not enforced
Moderation before public release	Needs stronger evidence

4. HLS and video-processing uncertainty

There is conflicting evidence in the reports.

The July audit says a server-side Storage trigger processes uploaded videos.

YohPal Live — Production Readiness Review.pdf

However, an earlier detailed pipeline audit found that:

- The Cloud Run transcoder returned only a queued response.

- The FFmpeg pipeline was skeletal.
- Uploaded raw MP4 files were temporarily written into `hlsUrl`.
- True adaptive HLS output was not yet implemented at that snapshot.

YohPal Mobile Flutter — Gap Report v10 — Video Pipeline Fixed + Chat Bugs Found.pdf

Therefore, developers must prove the current pipeline outputs real files such as:

```
master.m3u8
playlist_360p.m3u8
playlist_720p.m3u8
playlist_1080p.m3u8
thumbnail.webp
preview.mp4
segment_000.ts or .m4s
```

Until those files and playback manifests are shown from the latest candidate, HLS transcoding should be classified as:

UNVERIFIED, not certified.

5. My recommended disruptive launch position

YohPal should not try to clone every CapCut feature before launch.

The best launch strategy is:

YohPal AI Creator Copilot

Build an editor that helps ordinary users create professionally structured, locally relevant and commercially actionable videos in one guided workflow.

The differentiator should not merely be filters.

It should be:

YohPal understands what the creator is trying to achieve and builds the most effective version for Live, Jobs, Hustle, Market, education or business.

6. Recommended disruptive launch features

Launch Priority 1 — One-Tap Smart Edit

A user records or uploads raw footage and selects the objective:

- Grow followers
- Sell a product
- Advertise a job
- Promote a hustle
- Announce an event
- Teach something
- Share news
- Promote a course
- Build a business profile

YohPal should then automatically produce:

- Recommended trim
- Stronger first three seconds
- Auto captions
- Text overlays
- Suggested title
- Hashtags
- Thumbnail
- Music recommendation
- Call to action
- Product/job/course link
- Viral-readiness score
- Accessibility description

This is more disruptive than a generic editor because it connects creation directly to the rest of the ecosystem.

Launch Priority 2 — Real Native Timeline Editor

This is a launch-essential foundation.

Minimum capabilities:

- Trim start/end
- Split clips
- Delete segment
- Reorder clips
- Merge clips
- Crop to 9:16
- Rotate
- Volume
- Mute original audio
- Speed: 0.5x, 1x, 1.5x, 2x
- Undo/redo
- Before/after preview
- Save draft
- Render progress
- Export failure recovery

Without these, users must leave YohPal, edit elsewhere and return.

Launch Priority 3 — YohPal Camera

Add a first-class recording experience:

- Front and rear camera
- Flash
- 3-second and 10-second countdown
- Record in segments
- Pause/resume
- Delete last clip
- Speed control
- Timer
- Hands-free recording
- Grid and alignment
- Beauty controls
- Live captions while recording
- Teleprompter
- Product/job/course prompt cards

Disruptive feature

AI Director Mode

Before recording, the creator describes the video. YohPal generates:

- Shot sequence
 - Script
 - Teleprompter text
 - Scene duration
 - Camera suggestions
 - Props/product cues
 - Call-to-action ending
-

Launch Priority 4 — Automatic Spoken Captions

The editor should generate time-synchronized subtitles, not only a post caption.

Capabilities:

- Speech-to-text
- Word-level timing
- Caption styles
- Highlight current word
- Multiple languages
- English–Kiswahili translation
- African language expansion
- Editable transcript
- Burned-in or selectable subtitles
- Speaker identification
- Profanity masking

This can become a major African-market advantage.

Launch Priority 5 — Smart Music and Beat Editing

Add:

- Licensed music catalogue
- Original sounds
- Saved sounds
- Trending sounds
- Genre/mood selection
- Beat detection
- Auto-cut clips to beat
- Volume ducking under speech
- Fade in/out
- Voice-over
- Sound effects

YohPal should not launch music functionality without licensing and rights-management controls.

Launch Priority 6 — Ecosystem-Native Video Objects

This is where YohPal can outperform conventional short-video apps.

Creators should be able to attach interactive objects directly to the video timeline:

- Job vacancy
- Hustle task
- Market product
- Course/programme
- Coupon
- Poll
- Donation/gift action
- Business profile
- Event
- Live session
- AI Ask button
- Chat action

The object should appear at a chosen timestamp.

Example:

00:03 — Product card appears
00:08 — “Apply for this Job” appears
00:14 — Poll appears
00:20 — “Ask YohPal Brain” appears

This turns videos from passive content into actionable economic journeys.

Launch Priority 7 — Remix, Duet and Stitch

Add:

- Duet side-by-side
- Stitch selected segment
- Reaction recording
- Quote video
- Green-screen response
- Creator attribution
- Original-content link
- Permissions per video
- Revenue attribution to source creator

This strengthens network effects and creator collaboration.

Launch Priority 8 — AI Compliance and Safety Assistant

Before publishing, show:

- Copyright-risk warning
- Possible harmful-content warning
- Personal-information warning
- Child-safety review

- Deceptive-financial-claim warning
- Medical/legal-claim caution
- Brand/logo rights warning
- Product-price consistency check

The assistant should explain the issue and offer a safe correction.

7. Recommended launch scope

Must be ready at launch

1. Camera recording and gallery import
2. Native trim/split/reorder editor
3. Draft saving
4. Resumable upload with cancel and retry
5. Real HLS/MP4 processing proof
6. Auto spoken captions
7. Caption, hashtag, hook and thumbnail AI
8. AI checklist that genuinely gates publication
9. Music/voice-over basics
10. Product, Job, Hustle and Course attachment
11. Copyright and safety checks
12. Android and iPhone device certification

Controlled pilot after launch

- Beat-synchronized auto-edit
 - AI Director
 - Duet/stitch
 - Background removal
 - Generative video effects
 - AI avatar
 - Voice cloning
 - Scene replacement
 - Advanced beauty effects
-

8. Required codebase foundation

Canonical edit-project contract

```
enum CreatorObjective {
    entertain,
    growAudience,
    sellProduct,
    advertiseJob,
    promoteHustle,
    teach,
    promoteCourse,
    announceEvent,
}

final class VideoEditProject {
    const VideoEditProject({
        required this.id,
        required this.ownerId,
        required this.objective,
        required this.clips,
        required this.overlays,
        required this.audioTracks,
        required this.captionTracks,
        required this.status,
    });

    final String id;
    final String ownerId;
    final CreatorObjective objective;
    final List<VideoClipEdit> clips;
    final List<TimelineOverlay> overlays;
    final List<AudioTrackEdit> audioTracks;
    final List<CaptionTrack> captionTracks;
    final EditProjectStatus status;
}

final class VideoClipEdit {
```

```

const VideoClipEdit({
  required this.assetId,
  required this.sourcePath,
  required this.trimStartMs,
  required this.trimEndMs,
  required this.speed,
  required this.volume,
  required this.rotationDegrees,
});

final String assetId;
final String sourcePath;
final int trimStartMs;
final int trimEndMs;
final double speed;
final double volume;
final int rotationDegrees;
}

enum EditProjectStatus {
  draft,
  rendering,
  rendered,
  uploadQueued,
  uploading,
  processing,
  moderationPending,
  readyToPublish,
  published,
  failed,
}

```

Publication gate

```

final class VideoPublicationGate {
  const VideoPublicationGate();

  PublicationDecision evaluate({
    required bool renderComplete,
    required bool uploadComplete,
    required bool moderationPassed,
    required bool copyrightCheckPassed,
    required bool captionReviewComplete,
    required bool requiredMetadataComplete,
    required bool aiChecklistRequired,
    required bool aiChecklistComplete,
  }) {
    final blockers = <String>[];

    if (!renderComplete) {
      blockers.add('Video rendering is incomplete.');
```

```

    }

    if (!uploadComplete) {
      blockers.add('Video upload is incomplete.');
```

```

    }

    if (!moderationPassed) {
      blockers.add('Safety review has not passed.');
```

```

    }

    if (!copyrightCheckPassed) {
      blockers.add('Copyright review has not passed.');
```

```

    }

    if (!captionReviewComplete) {
      blockers.add('Caption review is incomplete.');
```

```

    }

    if (!requiredMetadataComplete) {
      blockers.add('Required publishing information is missing.');
```

```

    }

    if (aiChecklistRequired && !aiChecklistComplete) {
      blockers.add('AI publishing checklist is incomplete.');
```

```

    }

    return PublicationDecision(
      allowed: blockers.isEmpty,
      blockers: blockers,
    );
  }
}

final class PublicationDecision {
  const PublicationDecision({

```

```
        required this.allowed,
        required this.blockers,
    });

    final bool allowed;
    final List<String> blockers;
}
```

Upload reliability contract

```
abstract interface class ResumableVideoUploadService {
    Stream<VideoUploadState> watch(String uploadId);

    Future<String> start({
        required String projectId,
        required String renderedFilePath,
    });

    Future<void> pause(String uploadId);

    Future<void> resume(String uploadId);

    Future<void> cancel(String uploadId);

    Future<void> retry(String uploadId);
}
```

Recommended decision

Upload pipeline

Functional but not yet launch-differentiating.

AI publishing assistant

Promising and one of YohPal's strongest existing creator capabilities.

Native video editing

Insufficient for a disruptive short-video launch.

Marketing position today

Do not claim:

- Full AI video editor
- Automatic professional editing
- AI auto-trim execution
- Advanced music studio
- Complete HLS processing
- CapCut-equivalent creation suite

until device and output evidence proves those capabilities.

The honest launch language can be:

"AI-assisted publishing with captions, hooks, hashtags, thumbnail ideas, trim recommendations and content-readiness guidance."

Next Smart Move

Build YOHPAL LIVE CREATOR STUDIO RELEASE 1.1A — DISRUPTIVE VIDEO CREATION FOUNDATION, limited to a real YohPal Camera, multi-clip timeline editing, trim/split/reorder, synchronized auto-captions, music and voice-over, resumable uploads, enforced publication gates, and timestamped Jobs/Hustle/Market/Course objects. Retain the existing AI captions, hooks, hashtags, viral score, thumbnail ideas and trim suggestions, but connect them to actual media edits and verified YohPal Brain production responses.